The Personalist Penalty: Varieties of Autocracy and Economic Growth

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Abstract

We explore the economic consequences of personalist rule, where power is concentrated in an individual or small elite. Working with a cross-country panel from 1960 to 2010, and building on the empirical strategy of Acemoglu, Naidu, Restrepo, and Robinson (2019), we compare the growth performance of personalist autocracies, institutionalized autocracies, and democracies. Consistent with prior research, authoritarian regimes of all types have lower average growth rates than democratic ones, but this "autocratic penaltyâĂŹâĂŹ is largely concentrated in personalist autocracies. Exploring mechanisms, we find that autocracies with highly concentrated power have significantly lower private investment, more domestic unrest and protest, and lower total factor productivity. Our results highlight the very different incentives facing leaders in authoritarian regimes with narrow and broad bases of power.

1 Introduction

How do political institutions shape policy choices and economic performance? A decades-long literature has focused on comparing income growth in democracies versus autocracies.[1] The challenge with such comparisons, however, is that non-democratic institutions are far more diverse than democratic ones. For instance, "autocracy" pools institutionalized regimes led by Lee Kwan Yu (Singapore) or Chung Hee Park (South Korea) with personalist strongmen such as Mobutu Sese Seko (Zaire) or Saddam Hussein (Iraq).[2]

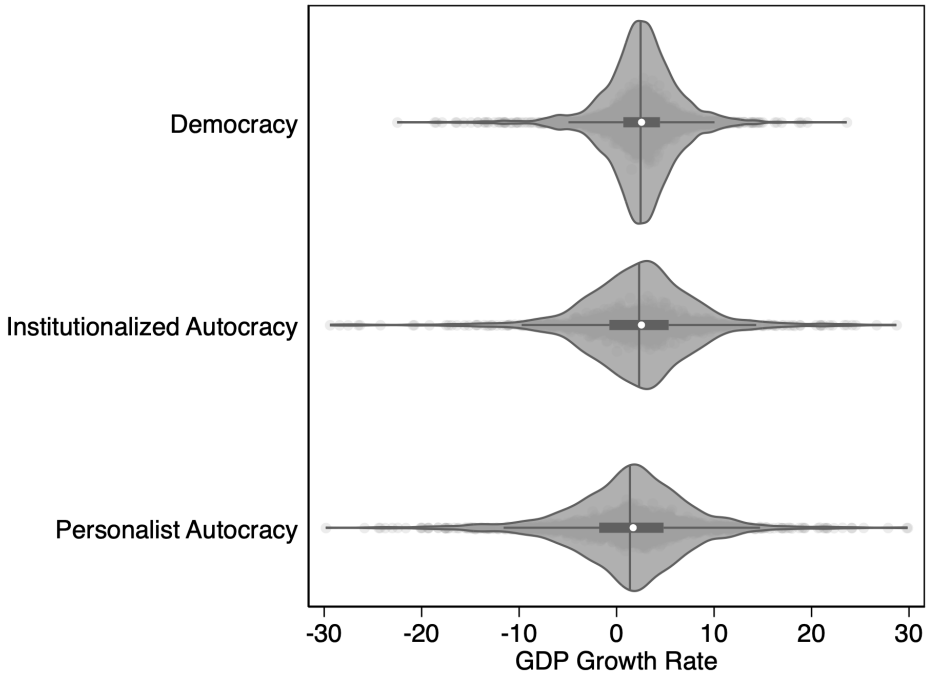
This paper focuses on the consequences of personalist rule, where power is concentrated in an individual or small elite. Currently, more than thirty autocratic governments are personalist dictatorships (Frantz, 2024). Two of the world's most powerful autocrats—Vladimir Putin of Russia and Xi Jinping of China—have been removing institutional checks and balances and ruling by personal influence and decree. The risks of such power consolidation is one of the most pressing questions for global political stability and economic development.

Compared to institutionalized autocrats, personalist rulers are less encumbered by the checks and balances that hold government accountable, limit the ability to make sweeping changes, and remove poor performers from office. In addition to this absence of accountability, personalist institutions often deliver incomplete and biased information to rulers (Wintrobe, 1998*b*; Gehlbach, Sonin and Svolik, 2016; Egorov and Sonin, 2024). As a consequence, many social scientists hypothesize that unconstrained rule leads to poorer policy and decision-making (Ostrom, 1997, 2010; Padró i Miquel, 2007; Besley and Kudamatsu, 2007).

[1] Generally speaking, early contributions failed to find a robust difference in growth rates between between the two, and sometimes found a weak adverse effect of democracy (Barro, 1996; Przeworski and Limongi, 1993; Gerring et al., 2005). More recent work using panel data methods, however, has argued that democratic transitions are associated with accelerations in growth, while autocratic backslides inhibits development (Rodrik and Wacziarg, 2005; Persson and Tabellini, 2006; Papaioannou and Siourounis, 2008; Wright, 2008; Acemoglu et al., 2019). We return to this literature and re-estimate using various measures of democracy and autocracy below.

[2] Concentration of power is just one way that autocracies vary. The category also encompasses a wide range of regime types—monarchies, military juntas, single party rule, absolutist dictatorships, and multiparty systems where a hegemonic party limits competition (Geddes, 1999). Moreover, formal institutions cannot be taken at face value in nondemocracies, where unwritten rules, power, and coercion determine political control (Gehlbach, Sonin and Svolik, 2016).

Figure 1: Mean and distribution of growth rates by regime type



Notes: The vertical line represents the median and the white dot the mean. The horizontal bar spans the 25th to 75th percentiles, and the horizontal line the 5th to 95th percentiles. This figure uses annual GDP growth rates from the October 2023 World Development Indicators (WDI). We classify regime types using the measures of autocracy and power consolidation by Cheibub, Gandhi and Vreeland (2010) and Gandhi and Sumner (2020). Sections 3.1 and 4 describe these data and choices in more detail.

There is some evidence that personalist dictators are more likely to stoke internal and external political conflict (Jackson and Morelli, 2007; Weeks, 2012; Blattman, 2022). More broadly, authoritarian rule is associated with lower investment in public goods and lower private investment (Bueno De Mesquita et al., 2003; Besley and Kudamatsu, 2007; Wright, 2008; Gehlbach and Keefer, 2011, 2012). It is an open question whether these adverse effects are driven by the autocracies where power is most concentrated.

In other words, the variation within nondemocratic regimes could matter as much or more than the autocratic–democratic distinction. Figure 1 plots annual growth rates from 1960 to 2010 in 175 countries by regime type: democracies, personalized autocracies, and institutionalized autocracies. There are at least eight measures of autocracy and six potential indicators of power concentration. Such classifications are inevitably subjective, each with

advantages and disadvantages, and this paper analyzes all the available datasets. Figure 1 employs a measure of power consolidation by Gandhi and Sumner (2020) that arguably best captures the concept of personalism and concentrated power (and is also the most complete, objective, and explicitly coded).

In this simple comparison, both types of autocracy have substantially higher variance. But personalized autocracies stand out for their lower per capita growth rates: an average of 2.4 for democracies, 2.31 for institutionalized autocracies, and 1.37 for personalized autocracies. This difference may seem subtle against the huge spread of growth rates, but it is substantial—compounded over 50 years, a 1 percentage point greater growth rate corresponds to 63% higher incomes.

The question, of course, is whether these differences are statistically significant and causal. Obviously, when it comes to estimating the effect of institutional arrangements on country development, there is no perfect strategy. Our preferred approach takes a panel of countries and regresses log income per capita on indicators for autocracy and personalist autocracies, as well as country and year fixed effects. This approach follows a recent democracy–autocracy comparison by Acemoglu et al. (2019) and, like them, we control for pre-transition dynamics with four lags of the dependent variable. The coefficients in this dynamic two-way fixed effects (TWFE) model have a relatively simple interpretation: they give us the conditional mean difference in growth rates between regime types. The coefficient on *autocracy* indicates the average difference between democratic and institutionalized autocracies, and the coefficient on the *personalism* indicator estimates the growth penalty from high levels of power concentration. These estimates are interpretable as the causal effect of transitions from one regime type to another under the additional assumption of "sequential exogeneity"—the typical assumption in dynamic panel models.

We explore a variety of empirical models and measures and most paint a picture similar to that in Figure 1: autocratic regimes of all types have lower average growth rates than democratic ones, but this adverse growth performance is largely concentrated in person-

alist autocracies. There is no robust difference in growth rates between democracies and institutionalized autocracies.

To investigate why, we collect a range of other political and economic indicators to understand the source of the personalist growth penalty. We see no robust evidence that personalism is associated with public investment, human capital investment, market openness, or civil war. Autocracies with highly concentrated power have more domestic unrest and protest, however, as well as significantly lower private investment and total factor productivity.

We are not the first to note that some autocracies are high-performing (Easterly, 2011; Luo and Przeworski, 2019). Indeed, some of history's most famous episodes of development happened under nondemocratic regimes—from Britain's industrial revolution to the "miracle" decades of growth in China, South Korea, and Taiwan (Besley and Kudamatsu, 2007). This is consistent with the view that, given appropriate incentives and a long enough time-horizon, autocrats will internalize the benefits of growth- and tax-maximizing policies and govern benevolently (Olson, 1993; Besley and Kudamatsu, 2007). Arguably, such assumptions do not hold for personalist dictatorships, and this could account for their underperformance.

Our analysis also builds on long-running efforts to unpack regime characteristics in nondemocracies (Geddes, 1999; Hadenius and Teorell, 2007; Gandhi, 2008; Goemans, Gleditsch and Chiozza, 2009; Cheibub, Gandhi and Vreeland, 2010; Svolik, 2012). Much of the empirical work has focused on the effects of personalism on political repression, conflict, and collapse (Choi, 2010; Weeks, 2012; Chin, Song and Wright, 2023; Frantz et al., 2020). Gandhi (2008) and Wright (2008) do examine differential growth rates in autocracies with and without formal institutions, but since that early work there has been surprisingly little research on how the concentration of power in autocracies shapes growth and development. That is the question we take up in this paper.

2 Conceptual framework

Many theories link political institutions to economic performance. To organize these, consider the classic augmented production function (Mankiw, Romer and Weil, 1992):

$$Y = K^{\alpha} H^{\beta} (AL)^{1-\alpha-\beta}.$$

Understood from this perspective, the concentration of political power and strength of formal institutions in autocratic regimes may affect growth through one or more of the four inputs into production: physical capital (K), human capital (H), technology (A), and labor (L).

A familiar argument is that dictators with a narrow base of power may prioritize targeted transfers over public goods (Bueno De Mesquita et al., 2003; Besley and Kudamatsu, 2007). To the extent this is the case, economic growth can suffer from underinvestment in human capital and technology, broadly defined.

Highly concentrated political power may also discourage private investment in physical capital. Gehlbach and Keefer (2011, 2012) argue that competitively elected legislatures and institutionalized ruling parties deter autocrats from expropriating politically connected elites. Institutionalized parties and legislatures also encourage policy stability (Gandhi, 2008)—a further precondition for private investment. Such constraints are weak or absent in personalist regimes, where "parties" or "legislatures" are enfeebled, functioning primarily to reward and punish elites, or absent entirely (Wright, 2008).

Personalized rulers may also be more likely to provoke civil unrest and interstate and civil war. Although there is always some misalignment of preferences between leaders and citizens, unconstrained autocrats internalize fewer of the costs, and potentially more of the benefits, of conflict (Bueno De Mesquita et al., 2003; Jackson and Morelli, 2007; Weeks, 2012; Blattman, 2022). Personalized institutions may also struggle to maintain stable political bargains. The absence or impotence of formal institutions in autocracies, for example, may complicate efforts to co-opt rivals before open conflict emerges (Gandhi and Przeworski, 2006; Svolik,

2012). Personalist dictators may also struggle to commit to rewarding those who protect them in times of crisis (Myerson, 2008).

Any such internal and external conflict has the potential to discourage growth. Wars destroy physical capital and reduce labor and human capital through schooling disruptions, casualties, conscription, and emigration (Blattman and Miguel, 2010). Although these factors can re-accumulate over time, the impact on national income tends to be large and persist for at least a decade (Mueller, 2012). Such effects may be most pronounced when accompanied by a deterioration in institutional quality—say, a reduction in constraints on executive behavior (Cerra and Saxena, 2008).

Finally, personalist autocrats may simply make worse policy decisions than their institutionalized counterparts—pursuing ill-conceived investment projects, for example, or invading a neighbor based on poor understanding of likely resistance. Wintrobe (1998*a*) suggests the presence of a "dictator's dilemma," in which self-censorship for fear of reprisal deprives the autocrat of information needed for his survival (see also Gehlbach et al., 2024). Although personalist dictatorships are not necessarily more repressive than are institutionalized autocracies, any tendency to tell the ruler what he wants to hear may be aggravated when the ruler makes decisions alone or in a small group. A related idea is that personalist autocrats may be more likely to surround themselves with loyal rather than competent subordinates (Egorov and Sonin, 2009; Zakharov, 2016; Egorov and Sonin, 2023). "Monocratic" rulers may also be more susceptible to influence from special interests (Gehlbach and Malesky, 2010), and they may be less flexible in their decision making than are actors in more "polycentric" systems (Vincent Ostrom, 1997; Elinor Ostrom, 2010).

We return to these alternative sources of any personalist penalty below in the discussion of measurement, results, and mechanisms, below.

3 Empirical strategy

3.1 Estimation

To examine growth differences by regime type, we take a panel of 175 countries from 1960 to 2010 in 175 countries and estimate a dynamic two-way fixed effects (TWFE) model:

$$y_{ct} = \beta_1 A_{ct} + \beta_2 P_{ct} + \sum_{j=1}^{J} \gamma_j y_{ct-j} + \alpha_c + \delta_t + \epsilon_{ct}. \tag{1}$$

Our dependent variable, y_{ct}, is GDP per capita for country c in year t. We have two main indicators of regime type: A_{ct} is a dichotomous measure of autocracy and P_{ct} is an indicator for personalist autocracies. They are not mutually exclusive categories, as P_{ct} is embedded in A_{ct}. α_c and δ_t are country and year fixed effects, and ϵ_{ct} is an idiosyncratic error term corrected for heteroskedasticity. The J lags of the dependent variable control for pre-transition income dynamics.

This TWFE approach is one of the most common ways to estimate growth differences in regime type.[3] This particular specification, including the lags of the dependent variable, follows from the Acemoglu et al. (2019) estimates of the effect of democracy on growth. Our estimate of the "autocratic penalty" is essentially the inverse of their specification, where $A = 0$ for democracies instead of 1:

$$y_{ct} = \beta A_{ct} + \sum_{j=1}^{J} \gamma_j y_{ct-j} + \alpha_c + \delta_t + \epsilon_{ct}. \tag{2}$$

In this simpler regression, the coefficient on autocracy estimates the average difference in growth rates between democratic and autocratic regimes. The literature mostly uses a dichotomous measure of autocracy, even though regime-type datasets typically measure democracy–autocracy on an ordinal scale. The chief reason for using an indicator is that it simplifies the dynamic panel analysis and identification requirements, as well as the interpre-

[3] NAME OTHERS WHO USE THIS GENERAL APPROACH

tation of the coefficient. At the same time, this simplification introduces measurement error. For instance, if income falls after autocratic shifts on the intensive margin, the coefficient on the autocracy indicator will be biased towards zero.

The advantage of adding a personalism indicator is that it accounts for an important dimension of heterogeneity in continuous autocracy scores. Once we introduce the personalist autocracy indicator in Equation 1, however, the interpretation of coefficients changes. β_1 estimates differences in log GDP per capita between autocracies and institutionalized (or non-personalist) autocracies, and β_2 measures differences between institutionalized and personalist autocracies. We again simplify continuous variation in personalism into a dichotomous variable for the same econometric and expositional reasons that most studies dichotomize autocracy.

3.2 Interpretation

Obviously, causal estimates from cross-national panel regressions must be treated with caution. At a minimum, the coefficients on autocracy and personalism have simple and clear descriptive interpretations—they estimate the conditional mean difference in log income per capita between regime types, adjusting for pre-transition income dynamics.

In addition, these coefficients can be interpreted as the "nearly identified" causal effect of a regime transition (in either direction). A key assumption is sequential exogeneity:

$$\mathbb{E}\left[\epsilon_{ct} \mid A_{ct} \ldots, A_{c1}, P_{ct} \ldots, P_{c1}, y_{ct-1}, \ldots, y_{c1}, \alpha_c, \delta_t, \ldots, \delta_1\right] = 0, \forall t \geq 1. \tag{3}$$

This assumption requires that countries undergoing a democratic transition or autocratic reversal are not on different national income trajectories than those that do not transition between regime types. It also requires that autocracies transitioning to or from personalist rule are not on different trajectories.

The plausibility of this assumption relies on having the "correct" lag structure, such that

the dynamics of GDP per capita are fully specified.[4] Acemoglu et al. (2019) found evidence of income losses prior to democratic transitions, suggesting it is important to control for these dynamics. They found that controlling for four lags appears to be sufficient for estimating Equation 2, and our analysis supports the same lag structure when personalism is included.[5]

Another identification issue is that the "assignment" of countries to autocratic or personalist rule is staggered, implying that our coefficient estimates are based in part on comparisons across earlier- and later-treated cohorts, with potentially negative weights on some comparisons (e.g,. Goodman-Bacon, 2021). Reversals (i.e., democratic transitions or shifts towards personalist rule) have the same implication. Recent methods to address this issue typically fail to anticipate the dynamics that Equations 1 and 2 model. Some fail to accommodate reversals, and none are appropriate for environments with multiple binary treatments (Roth et al., 2023; Chiu et al., 2023).[6]

Finally, we should interpret the coefficient on personalism as the effect of a *package* of institutional reforms and policy changes that commonly accompany a reduction in checks and balances on the leader. Indeed, the datasets we examine define personalism in such a multidimensional way. In terms of identification, however, we would be concerned if a set of non-personalist institutional changes tended to accompany switches in the personalist indicators. Examples include shifts in militarism or the general uncertainty and instability that can arise from a change in leader or regime. Where data are available, we can check for robustness to inclusion of these other changes in regime characteristics. But to the extent these or other unobserved autocratic characteristics are systematically related to a rollback

[4]Identification also relies on stationarity; following Acemoglu et al. (2019), we report p-values from a Levin-Lin-Chu test for unit roots. There remains "Nickell bias", which arises from the joint presence of lagged outcomes and unit fixed effects. As this is of order $1/T$, and our panel is 50 years long, any such bias is likely to be minimal—a conjecture supported by various analyses in Acemoglu et al. (2019).

[5]In a specification with eight lags, the hypothesis that lags 5–8 are jointly equal to zero cannot be rejected at conventional levels of significance.

[6]A partial exception is Imai, Kim and Wang (2023), who re-examine Acemoglu et al. (2019) with a matching estimator similar in spirit to de Chaisemartin and D'Haultfœuille (2023). They show that the autocratic penalty is robust to correcting for the bias from inappropriate comparisons arising from staggered treatment, and that this effect is driven by authoritarian reversals rather than democratic transitions. Like the other methods, however, this breaks down with two treatments.

in checks and balances, we can also think if these as part of the personalism package and interpret the coefficient accordingly.

3.3 Choosing a national income measure

We use GDP per capita in constant 2015 U.S. dollars from the World Bank's World Development Indicators (WDI). Income per capita measures are widely used, but version and quality issues are seldom discussed. Our results are generally robust to alternative measures, but the strengths and weaknesses of these data are important to highlight.[7]

The World Bank typically takes data "as is" from national-account files shared by countries. Countries regularly revise their national accounts, including past estimates of GDP, population, and deflators. Thus, different versions of the WDI can vary from month to month in terms of estimates and missingness (Goes, 2023*a*,*b*). We do not impute missing periods of data. Instead, we use a single (June 2022) version of the WDI, and investigate robustness of our analysis to other WDI versions and to the use of GDP data from the Penn World Tables.

Also note that the national income accounts of many countries are weak (especially in poorer countries) and commonly-used international datasets often have missingness, errors, and inconsistencies (Deaton and Heston, 2010; Jerven, 2013; Goes, 2023*b*). National income figures can also be subject to tampering and politics, especially in authoritarian regimes (Martinez, 2022). Since autocratic tampering this is a potential confounder in any analysis of regime type and growth, we will examine whether there is evidence that personalist regimes tamper more, and whether any such manipulation could influence our results.

Finally, both the WDI and the Penn World Tables (PWT) also offer a purchasing power parity (PPP) adjusted version of income. Instead of using market exchange rates, PPP-adjusted measures convert income into international prices using data on the local prices of

[7] Appendix Table A7 replicates our main personalism results for different versions of the WDI as well as for the Penn World Tables (version 10.01), and demonstrates that our findings are little affected by the vagaries of national-income reporting. Similarly, Appendix Table A8 shows that our results are generally robust to using purchasing power parity (PPP)-adjusted GDP estimates from the same data sources, though with standard errors that are roughly twice as large as in Table A7, there is some loss of statistical significance.

Table 1: Income and Autocracy

	Outcome: Log GDP per capita							
	(1) CGV	(2) GWF	(3) FH	(4) Polity	(5) ANRR	(6) BMR	(7) PS	(8) V-Dem
Autocracy	-0.583^{**}	-0.008	-0.648^{**}	-0.202	-0.744^{***}	-0.554^{*}	-0.812^{**}	-0.269
	(0.287)	(0.276)	(0.258)	(0.260)	(0.244)	(0.285)	(0.337)	(0.277)
p-value, unit root test	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Observations	5710	5104	5454	5519	6193	5602	5574	5858
Countries	176	144	177	154	179	176	155	165
Country & year FE	✓	✓	✓	✓	✓	✓	✓	✓
Four GDP lags	✓	✓	✓	✓	✓	✓	✓	✓

goods and services in a single year. These estimates are useful for cross-sectional comparisons of income levels (since non-traded goods and services are typically cheaper in poor countries). Johnson et al. (2013) and Deaton and Heston (2010) raise several issues with PPP measures of national income, however, and recommend against their use in long panel analysis for several reasons: variability across versions; questionable comparability across countries and years; and poor quality of price data in poorer and smaller countries. In our analysis, it would also conflate changes in price indexes with changes in national income estimates. We will examine PPP-adjusted measures for robustness but do not use them for the main analysis.

4 Results

4.1 The autocratic penalty

To frame the discussion of personalism, we begin by re-estimating the standard autocratic penalty using eight common measures, each one described in detail below. Table 1 reports results from the dynamic panel model in Equation 2, adjusting for income dynamics and conditioning on common shocks and time-invariant country characteristics. For clarity we multiply the estimated coefficient on autocracy by 100.

Autocracy is associated with lower levels of log income with all eight indicators, although one is extremely close to zero (-0.08, in Column 2). The other seven estimates range from -0.156 to -0.723, and five are statistically significant at conventional levels. These are all "large" in the sense of having substantial long run effects on income per capita. For instance,

the estimate in the first column, -0.582, implies a decrease in GDP per capita of just over one-half percent in the year of transition to autocracy, compounding thereafter. These results are broadly consistent with Acemoglu et al. (2019) who also find the penalty to be significant with most democarcy measures but not all.

If the premise of this paper is correct—that autocracies are highly heterogeneous—then we might expect the autocracy–growth relationship to vary with different samples or definitions. Nonetheless, the results in Table 1 provide cautious support for the idea that autocratic regimes underperform democracies in economic growth.[8] [CAN WE DESCRIBE A JOINT TEST OF SIGNIFICANCE? IS THERE SOME OTHER WAY TO DO THIS OTHER THAN "SEEMINGLY UNRELATED REGRESSION"?]

The autocracy indicators vary in their sample coverage and coding choices. These coding differences include: their definition of a democracy; whether autocracy is a residual category; how continuous scores are dichotimized; how they treat years of political instability, state collapse, or war; the dating of democratic and autocratic transitions; whether they use objective or subjective coding, rules; and whether they relay on the aggregation of expert sources. Some important features of each dataset are as follows:

- **CGV** Cheibub, Gandhi and Vreeland (2010) employ a minimalist definition of democracy based on a set of relatively objective criteria set out by Przeworski et al. (2000): popular, multi-party elections for the chief executive and legislature; and prior alternation in power under current electoral rules.

- **GWF** Geddes, Wright and Frantz (2014) employ criteria that they describe as intentionally less minimalist and objective. Democracy is the residual category, unlike many other measures, and so they they code fewer autocratic episodes than others. They also have 10 to 20 percent more years of missing data, because [DO WE KNOW

[8] Sequential exogeneity requires the inclusion of sufficiently many lags of the dependent variable to fully capture GDP dynamics. Appendix Table A5 demonstrates that the pattern identified in Acemoglu et al. (2019) holds more generally. For every definition of autocracy that we employ, we cannot reject the hypothesis that the coefficients on four lags beyond the four included here are jointly equal to zero. In every case, the estimated coefficient on autocracy is similar to that reported in Table 1.

WHY]?

- **Polity5** Marshall and Gurr (2020) develop a 21-point scale ranging from -10 (hereditary monarchy) to +10 (consolidated democracy) based on submeasures of competitiveness in executive recruitment, constraints on executive authority, and degree of political competition. Following Acemoglu et al. (2019), autocracies have a score of 0 or less.

- **Freedom House** Freedom House (2023) score countries on a 0-100 scale for 10 political rights and 15 civil liberties. [HOW DO WE CODE? WE DON'T USE THEIR ELECTORAL AND LIBERAL DEMOCRACIES?]

- **ANRR** Acemoglu et al. (2019) aggregate information primarily from Freedom House and an earlier version of Polity. A country-year is autocratic in ANRR if either Freedom House records it as "not free" or Polity provides a zero or negative value. To maximize time and country coverage, they also draw from other sources and make some of their own coding decisions. This gives the dataset roughly 20% more coverage than GWF and 8% more than CGV in our sample period.

- **PS** Papaioannou and Siourounis (2008) is similar to ANRR in that it combines information from the same two datasets, as well as from various historical sources, though it does not allow for authoritarian reversals following transitions to democracy.

- **BMR** Boix, Miller and Rosato (2013) is, like CGV, based on the minimalist definition of democracy from Przeworski et al. (2000), though it drops the alternation rule and accommodates autocoups.

- **V-Dem** Coppedge et al. (2024) relies heavily on expert coding and focuses on distinguishing varieties of democractic governance ("polyarchy") using dozens of indicators in 15 main categories. It codes regimes as "complete" autocracies if there are no multiparty elections for the chief executive or the legislature, and as "electoral autocracies"

if there are de-jure multiparty elections, but they fail to pass a sufficient threshold for being free and fair. We code both as autocracies.

Appendix Figure **??** illustrates differences in coding for a selection of countries. TO DISCUSS]

The variables are relatively highly correlated, with correlation coefficients typically ranging from 0.73 to 0.95 (see Appendix Figure **??**). Different results do not necessarily arise from the largest differences in sample coverage or correlation. For example, if we compare the ANRR to the Polity datasets, they have a correlation of 0.94 and, when restricted to the same sample of countries and years, yield estimates of 0.XX and 0.YY. Hence differences in coding rules are clearly influential.

The focus of this paper is on heterogeneity in autocratic performance. Accounting for varieties of autocratic regime should, in principle, lead to more consistent estimates of the autocratic penalty.

4.2 The personalist penalty

Table 2 adds an indicator of power consolidation in autocracies and estimates Equation 1. The coefficient on autocracy now represents the difference in log GDP per capita between democracies and institutionalized autocracies, while the coefficient on our measures of personalism estimate differences between personalist and institutionalized autocracies. We report the linear combination at the base of the table, which represents the difference in growth performance between personalized autocracies and democracies.

We examine six alternative measures. For consistency in measurement and sample, we nest measures of personalism in their autocracy "twin": GS Power Consolidation with CGV Autocracy, GWF Personalism (categorical and continuous) with GWF Autocracy, etc. For POLCON Veto Players, which has no corresponding measure of autocracy, we use CGV Autocracy. We also report results in which we pair GS Power Consolidation with ANRR Autocracy. We elaborate on each measure below.

Table 2: Income, Autocracy, and Personalism

	Outcome: Log GDP per capita						
	(1) CGV	(2) ANRR	(3) GWF	(4) GWF	(5) Polity	(6) V-Dem	(7) CGV
Autocracy	−0.057	−0.194	0.458	0.054	0.093	0.308	0.011
	(0.385)	(0.361)	(0.447)	(0.356)	(0.319)	(0.330)	(0.345)
GS power consolidation	−0.806**	−0.918**					
	(0.372)	(0.406)					
GWF Categorical Personalism			−0.916*				
			(0.547)				
GWF Continuous Personalism				−0.140			
				(0.397)			
Executive Constraints					−0.126		
					(0.393)		
V-Dem Presidentialism						−0.831***	
						(0.284)	
Veto Players							−0.879**
							(0.344)
Autocracy + Personalism	−0.863***	−1.111***	−0.458	−0.086	−0.033	−0.523*	−0.868**
	(0.306)	(0.355)	(0.321)	(0.312)	(0.318)	(0.306)	(0.341)
p-value, unit root test	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Observations	5710	5940	5104	5104	5395	5858	5516
Countries	176	178	144	144	154	165	172
Country & year FE	✓	✓	✓	✓	✓	✓	✓
Four GDP lags	✓	✓	✓	✓	✓	✓	✓

Note: All institutional variables are defined as dummies, with measures of personalism nested in measures of autocracy. Measures of personalism are ordered such that higher values imply fewer constraints.

The results suggest that, to the extent there is poor growth performance in authoritarian regimes, it is concentrated in personalist autocracies. First, note that adding a personalism indicator tends to shrink the autocratic growth penalty towards zero, and in no case is the estimated autocracy coefficient significantly different from zero. Four of the six estimates coefficients flip sign and are now positive. We cannot reject the claim that institutionalized autocracies and democracies have similar growth performance.[9]

In contrast, the estimated coefficient on every measure of personalism is negative, with a magnitude that is large and statistically significant in five of seven cases. In these five, a hypothetical transition from democracy to personalist autocracy is associated with an instantaneous decline in GDP of nearly one percentage point. Generally speaking, these results are robust to a range of changes in the model: to excluding planned economies;[10] to censoring extreme changes in growth;[11] and to concerns that income measures are more easily manipulated by dictators;[12] Not is there evidence that our results are driven by leadership transitions in genera.[13]

There are two exceptions to the personalist penalty in Table 2: GWF autocracy / GWF Continuous Personalism (Column 4) and Polity autocracy / Executive Constraints (Column

[9]As in Table 1, the unit root test strongly supports the assumption of stationarity for all regressions reported in Table 2. Similarly, Table **??** shows that for all autocracy/personalism pairs we cannot reject the hypothesis that the coefficients on lags 5–8 are jointly equal to zero.

[10]See Tables A11 and Table A12 for the exlcusion of planned economy periods and also all countries ever featuring planned economies.

[11]Table A9 restricts the sample to observations in which the absolute value of growth in GDP per capita from the previous year is less than 30 percent. Relative to the results reported in Table 2, the estimated effect of the personalism dummy is qualitatively similar.

[12]To examine this possibility, we adopt the strategy of Martinez (2022), who demonstrates that the elasticity of GDP with respect to nighttime luminosity (NTL) is greater in dictatorships than in democracies, which he interprets as evidence that autocracies are likely to exaggerate economic output. In Table A10, we extend his analysis to allow for different elasticities in institutionalized and personalist autocracies. Looking across indicators, we find no consistent evidence that GDP growth is systematically underreported in personalist autocracies, as might be implied by the negative estimated effect of personalism reported above.

[13]Leadership turnover may also affect growth directly—say, because of the disruption of informal networks that encourage investment (Malesky and Samphantharak, 2008; Earle and Gehlbach, 2015). The results in Table A13, in which we control for leadership change in the current period, support this conjecture. The effect, however, is nearly orthogonal to the focus of this paper: regardless of how institutions are measured, both the magnitude and statistical significance of the personalism estimate are essentially unchanged from Table 2.

5). In each case, the estimated difference in growth performance between institutionalized and personalized autocracies is small and insignificant. Notably, the GWF and Polity measures were also exceptional in Table 1, where there was no significant relationship between autocracy and growth using these measures. Where measures of institutions fail to uncover an autocratic penalty, there is also no evidence of a personalist penalty.

[TO ME HERE IS WHERE THE SUBSEQUENT DISCUSSION PERHAPS OUGHT TO FOCUS MORE ON THE DIFFERENCES BETWEEN THE TWO ODD MEASURES OUT. I"M NOT SURE WHAT FORMAL ANALYSIS WE CAN PROVIDE. I STILL FIND THE POLITY RESULT SUSPICIOUS HOWEVER GIVEN THAT ANRR IS SUPPOSEDLY VERY SIMILAR.]

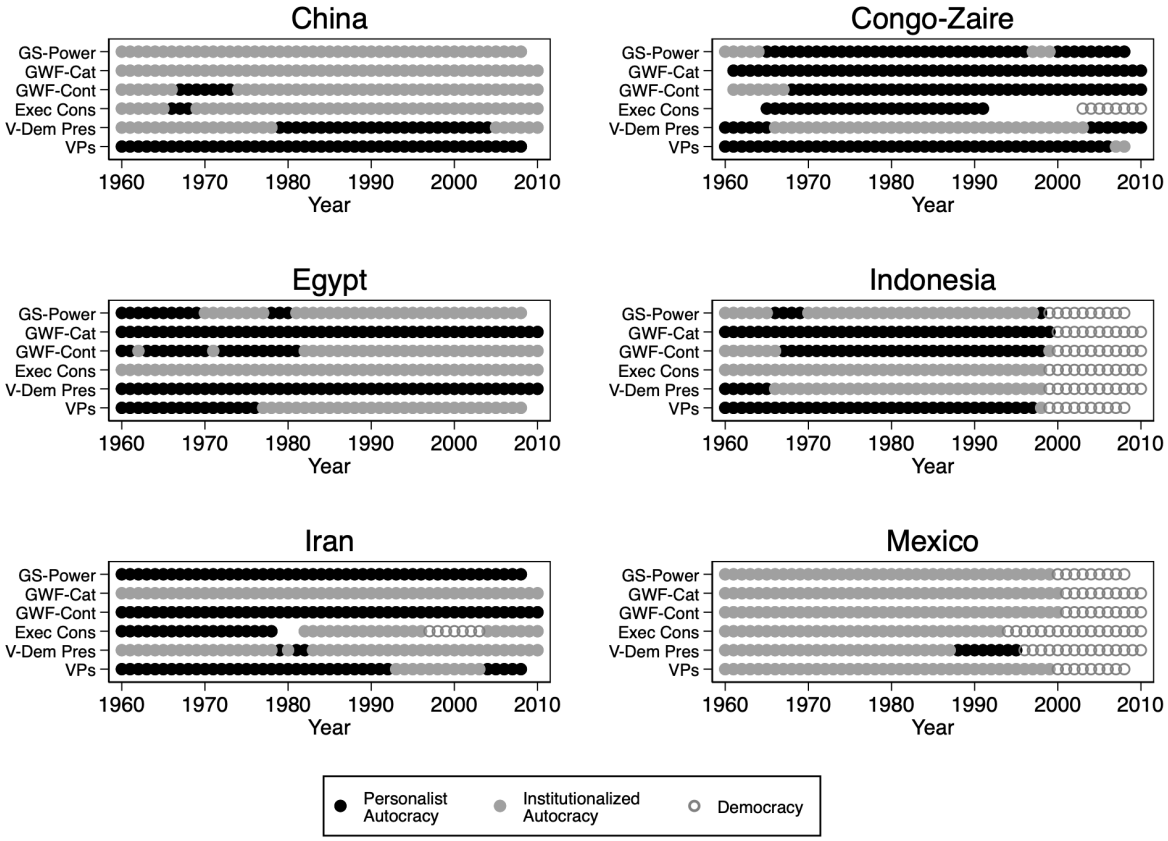
Figure 2 compares autocracy and personalism indicators for 6 countries, as examples. There is a moderate amount of agreement, but they nonetheless differ in important cases. In China, for instance, some code Maoist years as personalist and some not, while others code longer stretches as non-institutionalist. They also differ in how they code entire regimes, as in Congo-Zaire or Iran. Appendix Figures A1 through A6 illustrate the variation in the underlying continuous personalism data, along with specific leaders.

GS Power Consolidation Arguably closest to our framework is Gandhi and Sumner's (2020) measure of power consolidation in autocracies, which is estimated using variation in indicators of control over political office and freedom from military and party constraints.[14] GS index has several advantages. It is available for our full sample of 175 countries. It is time-varying within leader episodes, where the Bayesian item response theory (IRT) model on which it is based allows for smoothing across observations.[15] Finally, it is nested within an existing autocracy measure, CGV. Our personalism indicator equals one if an observation is an autocracy per CGV and the underlying IRT measure is above the median (zero).

[14]Examples of the former include the number of positions held by the ruler and whether any immediate family members hold political office, where examples of the latter include the presence and origins of parties associated with the ruler.

[15]In particular, for any period $t > 1$ within some leader episode $\{1, \ldots, T\}$, the prior distribution is set equal to the posterior distribution for period $t - 1$.

Figure 2: Alternative personalism codings in six representative countries



Notes: TO DO.

GWF Categorical and Continuous Personalism We also consider two measures of personalism by Geddes, Wright and Frantz (2014; 2018). Their 2014 work develops a categorical measure that codes autocracies into party-based, military, personalist, and monarchical autocracies (or hybrids). Our indicator equals one if GWF code a regime as personalist or hybrid personalist (e.g., military-personalist). The categorical measure has the advantage of being nested in their own autocracy measure, but has three disadvantages: there is no variation within leaders (meaning personalist transitions are "back dated"; the coding rules are comparatively subjective; and personalism coding is missing for roughly 20 percent of the 175 countries in our sample that are commonly coded as autocracies in other datasets.[16]

Geddes et al. (2018) also develop a continuous measure that, like Gandhi and Sumner (2020), is constructed with an IRT model. Their underlying various proxies include the ruler's connection to security forces, the relationship to the ruling party, and the role of the military.[17] The coding can also vary within leaders, though in this case the IRT model does not allow for smoothing within leader episodes. The coding is also still limited to the smaller number of GWF autocracies. Our indicator takes a value of one if the continuous measure is above the sample median, and zero otherwise. [REALLY WE SHOULD ENGAGE REASONS FOR DIVERGENCE HERE].

Polity5 Executive Constraints This variable attempts to measure the degree of institutionalized checks and balances on the executive, using one of the three scales used to construct the autocracy score. Executive constraints are coded on a 1 to 7 scale, with a value of 1 corresponding to "unlimited executive authority", and the codebook states that personalist dictatorships are typically coded here (Marshall and Gurr, 2020, p. 62). The variable is coded for the full sample, and can vary within leader, though in practice this is rare. The measure has several additional drawbacks. One is that a 1 on a scale of 7 is

[16]Because GWF code fewer autocracies than other datasets, there is effectively no GWF personalism coding for weak democracies that most ther datasets code as autocracies.

[17]Some of these indicators (e.g., whether the dictator created a new party to support the regime) are more objective than others (e.g., whether the party executive committee acts as a rubber stamp). Relative to Gandhi and Sumner (2020), there is more emphasis on coercive power, which receives less attention in the literature though may be consequential for economic growth.

intended to capture "unlimited authority" with no limitations on executive action, whereas as a 3 captures some limitations (such as moderate restraints by a legislature. As a result, non-personalist autocracies can only take the value of X to Y on the 1–7 scale. From Figure 2 we can see that this results in some significant departures from other measures, especially Iran, Egypt, and Indonesia. [REALLY WE SHOULD ENGAGE REASONS FOR DIVERGENCE HERE].

V-Dem Presidentialism The V-Dem variable `v2xnp_pres` attempts to measures the concentration of political power in a single individual. [SCOTT I THINK YOU DID NOT LIKE THIS VARIABLE BUT THE DESCRIPTION SEEMS FITTING YET THE PATTERNS ARE ODD. THOUGHTS?] We code this indicator based on the sample median for autocracies as coded by CGV.

POLCON Veto Players Henisz's (2000) measure of political constraints captures the number of actors whose agreement is necessary for policy change. We use the `POLCONV` version of this variable, identifying an autocratic leader (in CGV) as constrained if and only if the measure is above the sample median.

5 Mechanisms

Table 3: Potential mechanisms

Panel A: Economic Variables	Investment			Public Goods			Other Economic Variables	
	(1) Log Private Investment	(2) Log Public Investment	(3) Log TFP	(4) Log Primary Enrollment	(5) Log Secondary Enrollment	(6) Log Child Mortality Rate	(7) State Capacity	(8) Inflation Crisis
CGV Autocracy	0.001	−0.029	1.199***	−0.228	−0.886	0.089	−0.048***	0.012
	(0.023)	(0.022)	(0.354)	(0.430)	(0.858)	(0.089)	(0.010)	(0.014)
GS Power Consolidation	−0.039**	0.002	−1.487***	−0.726*	0.066	0.079	−0.007	0.001
	(0.017)	(0.019)	(0.476)	(0.401)	(0.691)	(0.082)	(0.010)	(0.009)
Autocracy + Personalism	−0.038*	−0.027	−0.288	−0.954**	−0.820	0.168**	−0.055***	0.013
	(0.020)	(0.022)	(0.367)	(0.404)	(0.656)	(0.069)	(0.011)	(0.013)
Mean of Outcome Variable	6.517	5.477	-3.477	452.903	393.629	352.929	0.285	0.071
Observations	5358	5358	3466	3394	2618	5499	5084	5710
Countries	163	163	106	167	158	174	156	176

Panel B: Conflict Variables	Civil Unrest		Civil Conflict			Civil War		
	(1) Log Protests	(2) Riots	(3) Incidence	(4) Onset	(5) End	(6) Incidence	(7) Onset	(8) End
CGV Autocracy	0.050	−0.001	0.013	0.003	−0.078	−0.006	0.007	−2.213**
	(0.069)	(0.030)	(0.018)	(0.017)	(0.083)	(0.014)	(0.007)	(0.799)
GS Power Consolidation	0.126	0.088***	−0.007	−0.005	0.038	0.009	0.002	1.809**
	(0.084)	(0.028)	(0.019)	(0.016)	(0.088)	(0.016)	(0.011)	(0.799)
Autocracy + Personalism	0.176**	0.087***	0.006	−0.002	−0.040	0.002	0.009	−0.404***
	(0.069)	(0.025)	(0.016)	(0.018)	(0.058)	(0.011)	(0.007)	(0.142)
Mean of Outcome Variable	-4.825	0.254	0.168	0.023	0.073	0.048	0.009	0.218
Observations	2802	5112	5377	3943	537	5377	4765	78
Countries	160	172	164	155	43	164	162	14

6 Conclusions

If you believe in the autocratic growth penalty, you should also believe that it is concentrated in personalist regimes.

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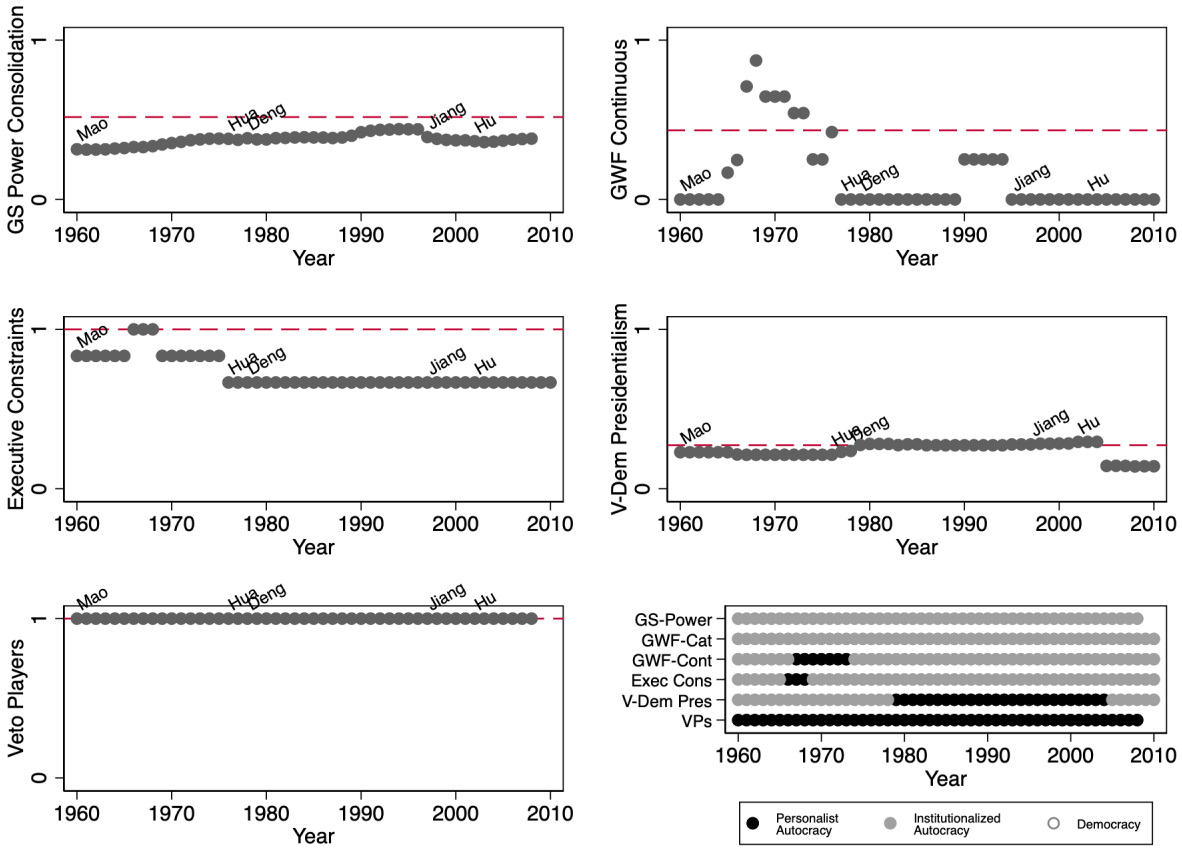
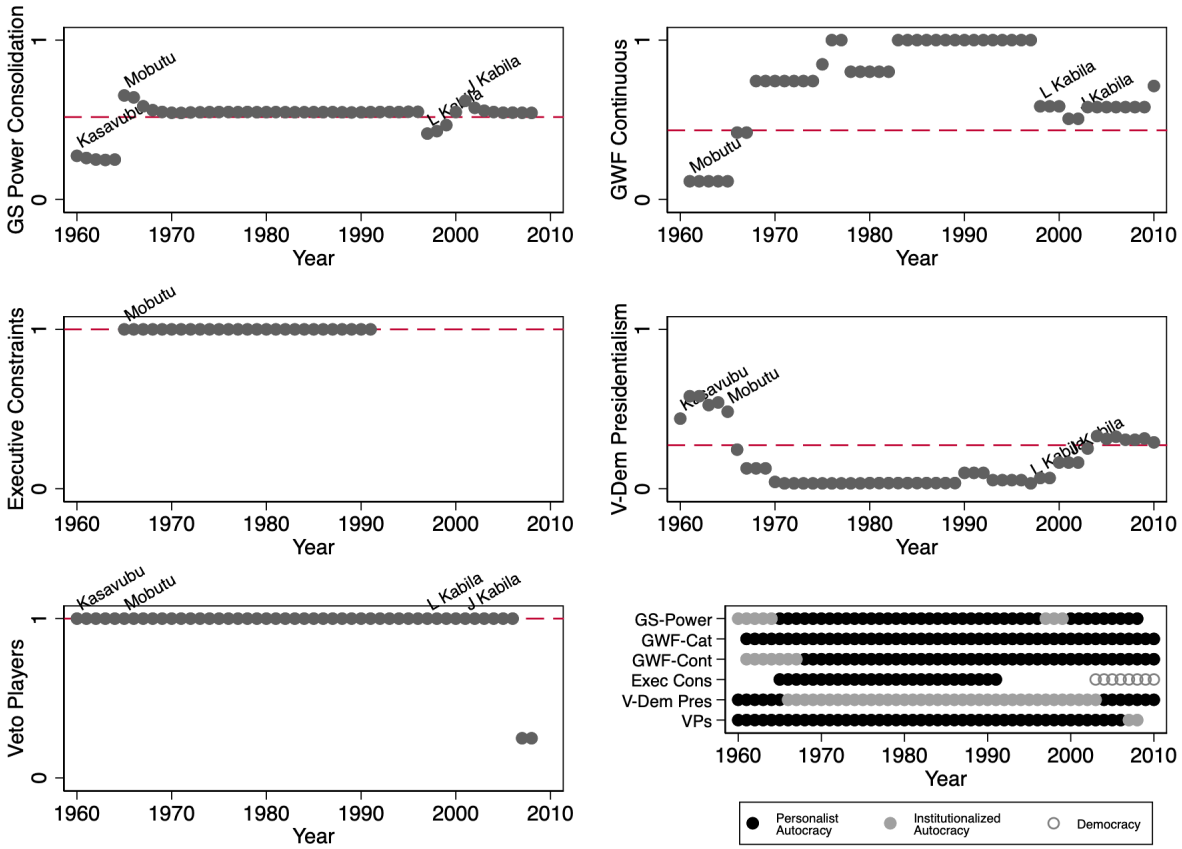
Appendix

Figure A1: Personalism measures for China



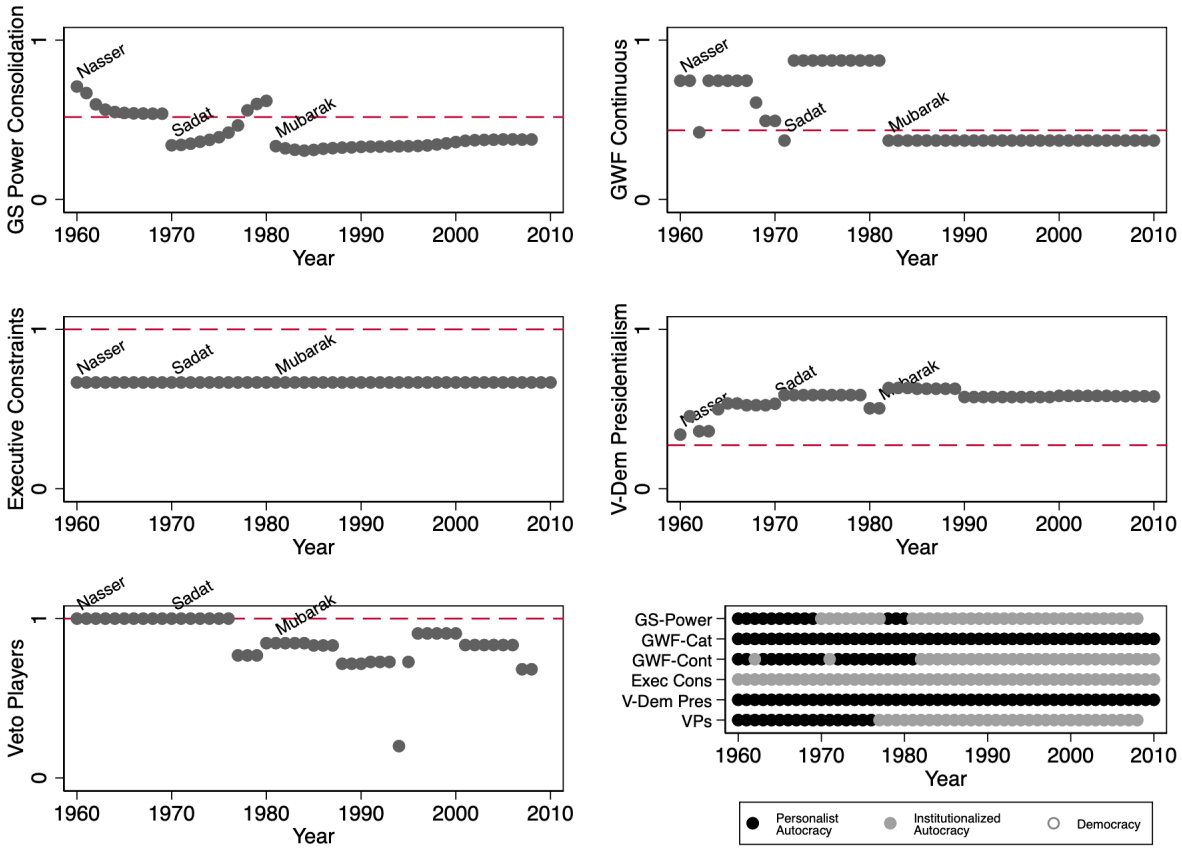
Notes: The Power Consolidation measure is normalized between its minimum and maximum values, yielding a score of 0.517, while Executive Constraint is scaled in reverse, resulting in a value of 1. For the remaining variables, the red dotted lines indicate their medians conditional on autocracies, and all these thresholds represented by the dotted red lines serve to dichotomize the personalistic measure.

Figure A2: Personalism measures for Congo-Zaire



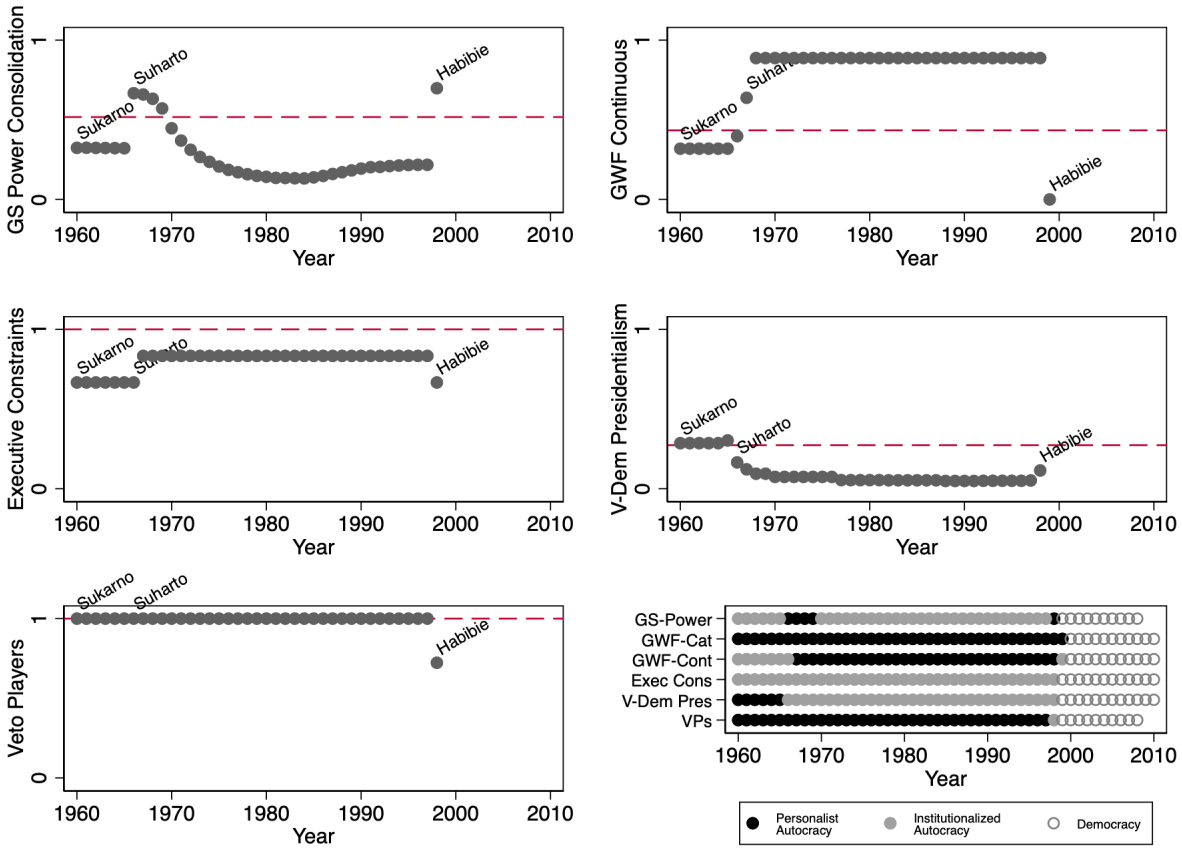
Notes: The Power Consolidation measure is normalized between its minimum and maximum values, yielding a score of 0.517, while Executive Constraint is scaled in reverse, resulting in a value of 1. For the remaining variables, the red dotted lines indicate their medians conditional on autocracies, and all these thresholds represented by the dotted red lines serve to dichotomize the personalistic measure.

Figure A3: Personalism measures for Egypt



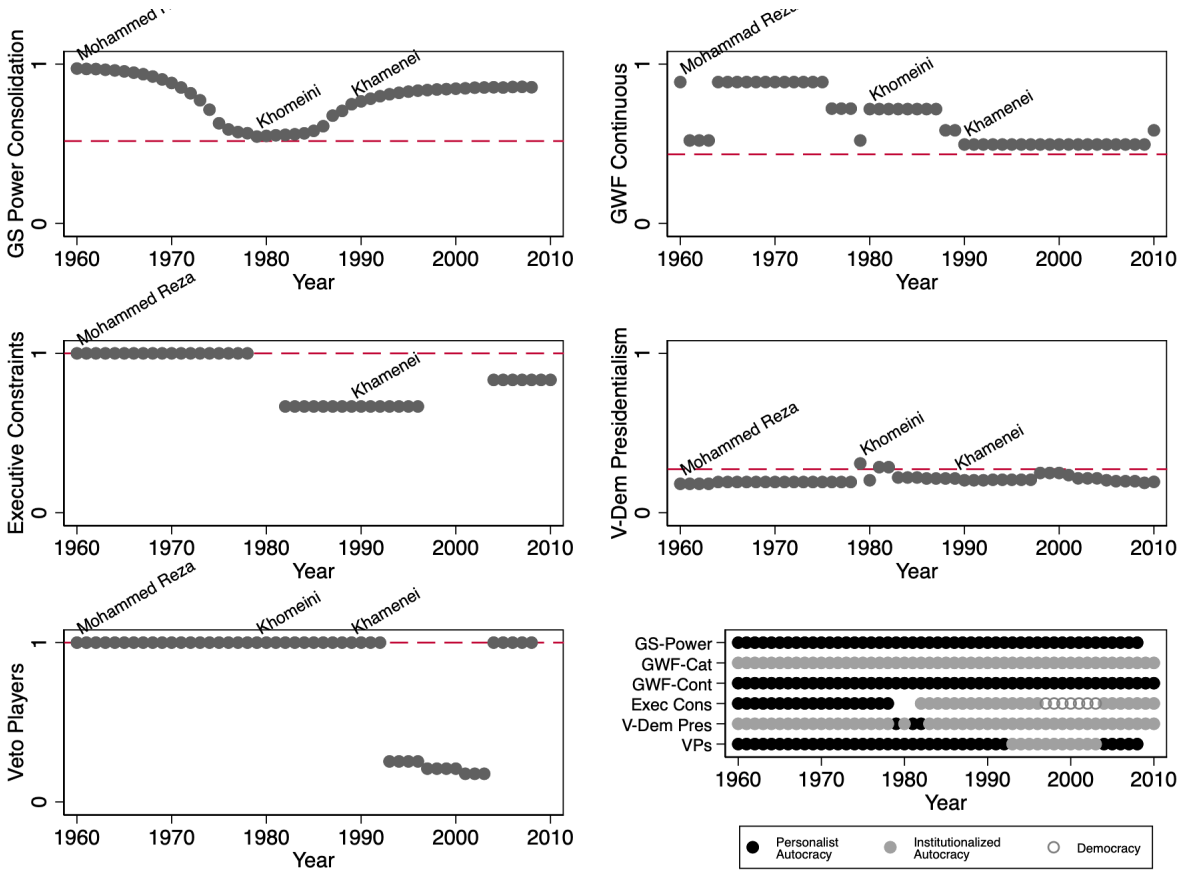
Notes: The Power Consolidation measure is normalized between its minimum and maximum values, yielding a score of 0.517, while Executive Constraint is scaled in reverse, resulting in a value of 1. For the remaining variables, the red dotted lines indicate their medians conditional on autocracies, and all these thresholds represented by the dotted red lines serve to dichotomize the personalistic measure.

Figure A4: Personalism measures for Indonesia



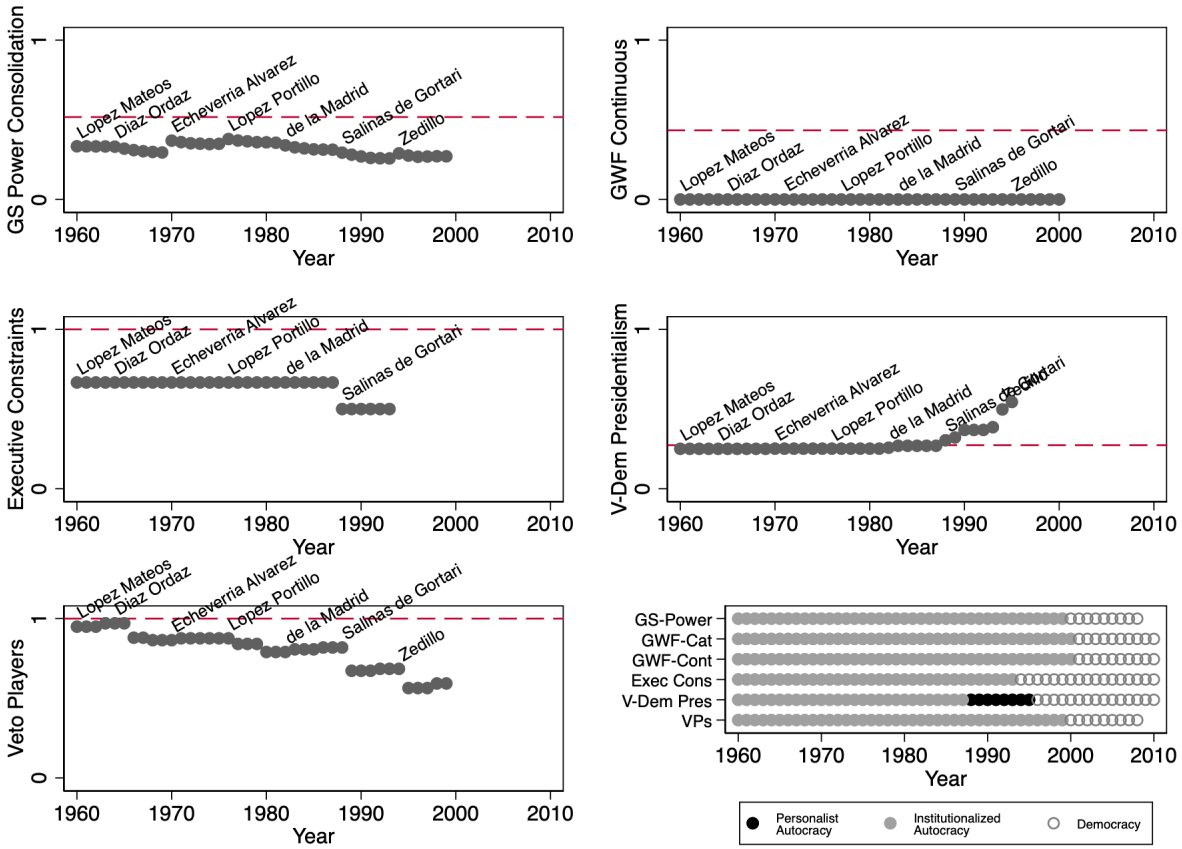
Notes: The Power Consolidation measure is normalized between its minimum and maximum values, yielding a score of 0.517, while Executive Constraint is scaled in reverse, resulting in a value of 1. For the remaining variables, the red dotted lines indicate their medians conditional on autocracies, and all these thresholds represented by the dotted red lines serve to dichotomize the personalistic measure.

Figure A5: Personalism measures for Iran



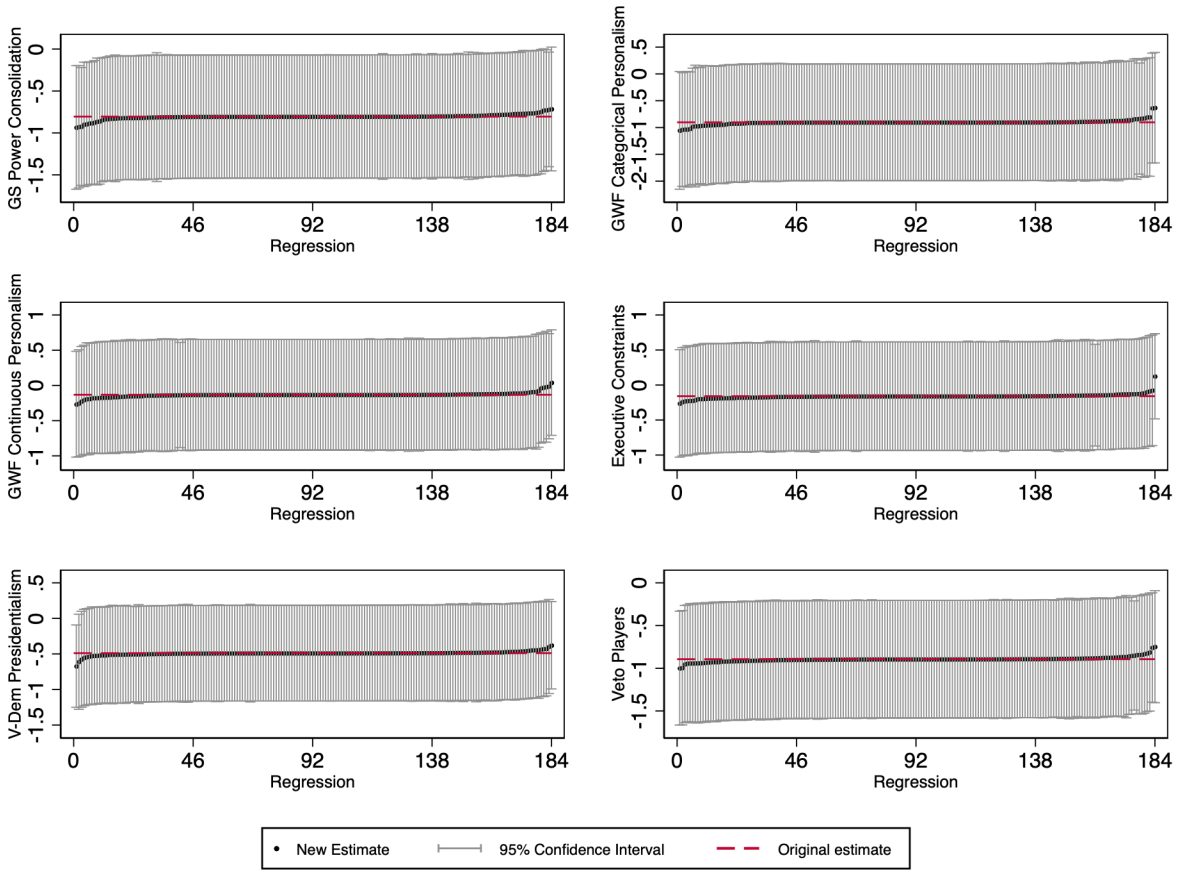
Notes: The Power Consolidation measure is normalized between its minimum and maximum values, yielding a score of 0.517, while Executive Constraint is scaled in reverse, resulting in a value of 1. For the remaining variables, the red dotted lines indicate their medians conditional on autocracies, and all these thresholds represented by the dotted red lines serve to dichotomize the personalistic measure.

Figure A6: Personalism measures for Mexico



Notes: The Power Consolidation measure is normalized between its minimum and maximum values, yielding a score of 0.517, while Executive Constraint is scaled in reverse, resulting in a value of 1. For the remaining variables, the red dotted lines indicate their medians conditional on autocracies, and all these thresholds represented by the dotted red lines serve to dichotomize the personalistic measure.

Figure A7: Leave-one-out personalism estimates at country level for the baseline growth regressions



Notes: In the regression figure, we present a visual representation of the estimated coefficients for all countries, with the specific country identified by its code being excluded from the analysis. The x-axis denotes the country codes, while the y-axis illustrates the estimated regression coefficients. The red dashed lines presents the original coefficients from 2. Each point corresponds to a coefficient derived from the regression model, reflecting the influence of the respective country on the overall outcome variable. This approach facilitates an understanding of how the absence of a particular country affects the overall model estimates, highlighting any potential outlier effects or influential data points. By visualizing these estimates, we provide insights into the robustness of the regression results.

Table A1: Measures of Power Consolidation in Autocratic Regimes

	GS Power Consolidation	GWF Categorical Personalism	GWF Continuous Personalism	Polity5 Executive Constraints	V-Dem Presi-dentialism	POLCON Veto Players
Conceptual match	High	Moderate	Moderate	Low	Low	Low
Country coverage	175	144	144	170	173	171
Varies within leader	Y	N	Y	Y	Y	Y
Objective coding	High	Moderate	Moderate	Low	Moderate	High
Nested in autocracy	Y	Y	Y	N	N	N

Note: Country coverage refers to regression sample in Table 2.

Table A2: Income and Autocracy with GWF Sample

	Outcome: Log GDP per capita							
	(1) CGV	(2) GWF	(3) FH	(4) Polity	(5) ANRR	(6) BMR	(7) PS	(8) V-Dem
Autocracy	−0.463	−0.005	−0.366	−0.142	−0.573**	−0.328	−0.831**	−0.107
	(0.301)	(0.276)	(0.283)	(0.247)	(0.268)	(0.318)	(0.392)	(0.288)
p-value, unit root test	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Observations	4749	5085	4412	5008	5085	4650	4668	5085
Countries	142	144	143	141	144	143	131	144
Country & year FE	✓	✓	✓	✓	✓	✓	✓	✓
Four GDP lags	✓	✓	✓	✓	✓	✓	✓	✓

Table A3: Income, Autocracy, and Personalism with GWF Sample

	Outcome: Log GDP per capita						
	(1) CGV	(2) ANRR	(3) GWF	(4) GWF	(5) Polity	(6) V-Dem	(7) CGV
Autocracy	0.101	−0.030	0.461	0.057	0.107	0.475	0.112
	(0.438)	(0.403)	(0.447)	(0.356)	(0.285)	(0.371)	(0.358)
GS power consolidation	−0.876*	−0.919*					
	(0.457)	(0.489)					
GWF Categorical Personalism			−0.916*				
			(0.547)				
GWF Continuous Personalism				−0.141			
				(0.396)			
Executive Constraints					0.092		
					(0.317)		
V-Dem Presidentialism						−0.894**	
						(0.373)	
Veto Players							−0.914**
							(0.372)
Autocracy + Personalism	−0.775**	−0.949**	−0.455	−0.084	0.200	−0.419	−0.802**
	(0.331)	(0.375)	(0.321)	(0.312)	(0.313)	(0.338)	(0.353)
p-value, unit root test	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Observations	4749	4923	5085	5085	4916	5085	4694
Countries	142	143	144	144	141	144	142
Country & year FE	✓	✓	✓	✓	✓	✓	✓
Four GDP lags	✓	✓	✓	✓	✓	✓	✓

Note: All institutional variables are defined as dummies, with measures of personalism nested in measures of autocracy. Measures of personalism are ordered such that higher values imply fewer constraints.

Table A4: Income, Autocracy, and Personalism with Polity Sample

	Outcome: Log GDP per capita						
	(1) CGV	(2) ANRR	(3) GWF	(4) GWF	(5) Polity	(6) V-Dem	(7) CGV
Autocracy	0.007	−0.120	0.485	0.074	0.152	0.333	0.070
	(0.375)	(0.357)	(0.450)	(0.359)	(0.318)	(0.327)	(0.344)
GS power consolidation	−0.723**	−0.798**					
	(0.358)	(0.399)					
GWF Categorical Personalism			−0.920*				
			(0.547)				
GWF Continuous Personalism				−0.131			
				(0.398)			
Executive Constraints					−0.159		
					(0.391)		
V-Dem Presidentialism						−0.765***	
						(0.271)	
Veto Players							−0.898**
							(0.362)
Autocracy + Personalism	−0.715**	−0.918***	−0.435	−0.057	−0.008	−0.432	−0.828**
	(0.300)	(0.352)	(0.319)	(0.312)	(0.319)	(0.302)	(0.342)
p-value, unit root test	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Observations	5075	5270	5008	5008	5349	5451	4992
Countries	152	153	141	141	153	153	152
Country & year FE	✓	✓	✓	✓	✓	✓	✓
Four GDP lags	✓	✓	✓	✓	✓	✓	✓

Note: All institutional variables are defined as dummies, with measures of personalism nested in measures of autocracy. Measures of personalism are ordered such that higher values imply fewer constraints.

Table A5: Income and Autocracy: Eight GDP Lags

	CGV	GWF	FH	Polity	ANRR	BMR	PS	V-Dem
Autocracy	-0.519*	0.073	-0.504*	-0.161	-0.755***	-0.544*	-0.869**	-0.273
	(0.310)	(0.293)	(0.277)	(0.296)	(0.266)	(0.292)	(0.374)	(0.276)
p-value, lags 5–8	0.313	0.259	0.552	0.259	0.423	0.221	0.201	0.391
Country & Year FE	Y	Y	Y	Y	Y	Y	Y	Y
Observations	5,038	4,557	5,090	4,896	5,490	4,924	4,924	5,165
Countries	173	144	175	153	177	171	154	164

Table A6: Income, Autocracy, and Personalism: Eight GDP Lags

	CGV	ANRR	GWF	GWF	CGV	CGV	CGV
Autocracy	-0.120	-0.374	0.671	0.184	-0.310	0.089	0.126
	(0.436)	(0.383)	(0.467)	(0.383)	(0.342)	(0.367)	(0.365)
GS Power Consolidation	-0.595	-0.733					
	(0.433)	(0.459)					
GWF Categorical Personalism			-1.139**				
			(0.557)				
GWF Continuous Personalism				-0.252			
				(0.413)			
Executive Constraints					0.329		
					(0.385)		
Veto Players						-0.883**	
						(0.383)	
V-Dem Presidentialism							-1.132***
							(0.368)
p-value, lags 5–8	0.338	0.234	0.252	0.266	0.414	0.281	0.335
Country & Year FE	Y	Y	Y	Y	Y	Y	Y
Observations	5,038	5,271	4,557	4,557	4,815	4,865	4,993
Countries	173	175	144	144	169	169	171

Note: All institutional variables are defined as dummies, with measures of personalism nested in measures of autocracy. Measures of personalism are ordered such that higher values imply fewer constraints.

Table A7: Robustness to Version of WDI GDP Data

	Outcome: Log GDP per capita						
	(1) April 2012	(2) April 2014	(3) February 2016	(4) February 2018	(5) February 2020	(6) February 2022	(7) Penn World Tables
CGV Autocracy	0.042	−0.018	−0.025	0.049	−0.095	−0.083	0.439
	(0.346)	(0.360)	(0.361)	(0.359)	(0.368)	(0.385)	(0.408)
GS Power Consolidation	−0.984***	−0.901***	−1.021***	−1.180***	−0.783**	−0.760**	−1.145***
	(0.341)	(0.342)	(0.369)	(0.364)	(0.368)	(0.372)	(0.401)
Observations	5870	5815	5776	5744	5816	5691	6312
Countries	175	176	176	176	176	176	168
Country & year FE	✓	✓	✓	✓	✓	✓	✓
Four GDP lags	✓	✓	✓	✓	✓	✓	✓

This table presents estimates of the effect of political regime on log GDP per capita. The reported coefficient on autocracy is multiplied by 100. In all specifications, we control country and year fixed effects. Cluster-robust standard errors that allow for intra-country correlation are presented in parenthesis. The seven columns use the following versions of WDI GDP data, which are the earliest version released by the World Bank in a given year: April 2012, April 2014, February 2016, February 2018, February 2020 and February 2022.

Table A8: Robustness to Version of WDI GDP Data (PPP)

	Outcome: Log GDP per capita (PPP)						
	(1) April 2012	(2) April 2014	(3) February 2016	(4) February 2018	(5) February 2020	(6) February 2022	(7) Penn World Tables
CGV Autocracy	−0.444	−0.561	−0.101	−0.186	0.337	0.533	0.867
	(0.571)	(0.634)	(0.599)	(0.623)	(0.531)	(0.628)	(0.591)
GS Power Consolidation	−0.301	−0.254	−0.892	−0.915	−1.289	−1.416	−1.924***
	(0.587)	(0.654)	(0.762)	(0.804)	(0.791)	(0.945)	(0.494)
Observations	3886	3807	2453	2480	2473	2447	6312
Countries	176	174	171	173	172	172	168
Country & year FE	✓	✓	✓	✓	✓	✓	✓
Four GDP lags	✓	✓	✓	✓	✓	✓	✓

This table presents estimates of the effect of political regime on log GDP per capita (PPP). The reported coefficient on autocracy is multiplied by 100. In all specifications, we control country and year fixed effects. Cluster-robust standard errors that allow for intra-country correlation are presented in parenthesis. The seven columns use the following versions of WDI GDP data, which are the earliest version released by the World Bank in a given year: April 2012, April 2014, February 2016, February 2018, February 2020 and February 2022.

Table A9: Omitting Extreme Values

	Outcome: Log GDP per capita						
	(1) CGV	(2) ANRR	(3) GWF	(4) GWF	(5) Polity	(6) V-Dem	(7) CGV
Autocracy	0.155	−0.011	0.384	0.209	0.034	0.253	0.136
	(0.331)	(0.306)	(0.379)	(0.299)	(0.275)	(0.314)	(0.277)
GS power consolidation	−0.870**	−0.893**					
	(0.341)	(0.353)					
GWF Categorical Personalism			−0.433				
			(0.478)				
GWF Continuous Personalism				−0.101			
				(0.328)			
Executive Constraints					−0.063		
					(0.298)		
V-Dem Presidentialism						−0.588**	
						(0.273)	
Veto Players							−0.769**
							(0.299)
Autocracy + Personalism	−0.716***	−0.905***	−0.049	0.108	−0.029	−0.334	−0.633**
	(0.276)	(0.274)	(0.284)	(0.266)	(0.290)	(0.274)	(0.307)
p-value, unit root test	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Observations	5660	5889	5077	5077	5343	5800	5469
Countries	175	177	144	144	153	164	171
Country & year FE	✓	✓	✓	✓	✓	✓	✓
Four GDP lags	✓	✓	✓	✓	✓	✓	✓

Only using the sample that has gdp per capita growth rate (in 2015 constant us dollars) falls between -30 to 30 percent. All institutional variables are defined as dummies, with measures of personalism nested in measures of autocracy. Measures of personalism are ordered such that higher values imply fewer constraints

Table A10: Differential misreporting of GDP

	Outcome: Log Total GDP						
	(1) CGV	(2) ANRR	(3) GWF	(4) GWF	(5) Polity	(6) V-Dem	(7) CGV
Autocracy	3.071	1.330	10.727***	6.354*	2.247	2.445	2.699
	(3.528)	(3.990)	(3.201)	(3.333)	(2.350)	(3.904)	(2.564)
Personalism	−1.317	−1.104	−10.989***	−5.491	−6.313**	−2.225	−2.727
	(4.373)	(5.061)	(3.864)	(5.278)	(3.146)	(3.628)	(2.685)
ln(NTL)	23.317***	22.284***	23.988***	24.161***	25.553***	26.407***	21.487***
	(4.489)	(4.257)	(3.732)	(3.785)	(3.611)	(3.824)	(4.181)
Autocracy × ln(NTL)	2.039	2.375	4.118**	3.345**	1.246	0.350	−0.879
	(2.006)	(1.717)	(1.678)	(1.448)	(1.211)	(1.929)	(1.416)
GS power consolidation × ln(NTL)	−1.538	−2.026					
	(2.108)	(1.956)					
GWF Categorical Personalism × ln(NTL)			−4.641**				
			(2.012)				
GWF Continuous Personalism × ln(NTL)				−4.579**			
				(2.309)			
Executive Constraints × ln(NTL)					−1.958		
					(1.528)		
V-Dem Presidentialism × ln(NTL)						0.573	
						(1.834)	
Veto Players × ln(NTL)							1.889**
							(0.903)
Observations	2059	2059	2059	2059	2059	2059	2059
Countries	166	166	166	166	166	166	166
Country & year FE	✓	✓	✓	✓	✓	✓	✓

Table A11: Excluding Planned Economies

	Outcome: Log GDP per capita						
	(1) CGV	(2) ANRR	(3) GWF	(4) GWF	(5) Polity	(6) V-Dem	(7) CGV
Autocracy	−0.147	−0.241	0.444	−0.040	−0.003	0.294	−0.037
	(0.390)	(0.365)	(0.461)	(0.361)	(0.319)	(0.329)	(0.343)
GS power consolidation	−0.750**	−0.862**					
	(0.377)	(0.403)					
GWF Categorical Personalism			−0.946*				
			(0.558)				
GWF Continuous Personalism				−0.018			
				(0.404)			
Executive Constraints					−0.010		
					(0.403)		
V-Dem Presidentialism						−0.862***	
						(0.284)	
Veto Players							−0.930***
							(0.343)
Autocracy + Personalism	−0.898***	−1.103***	−0.502	−0.058	−0.013	−0.568*	−0.967***
	(0.302)	(0.341)	(0.318)	(0.322)	(0.323)	(0.311)	(0.342)
p-value, unit root test	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Observations	5636	5868	5034	5034	5319	5779	5442
Countries	175	177	143	143	153	164	171
Country & year FE	✓	✓	✓	✓	✓	✓	✓
Four GDP lags	✓	✓	✓	✓	✓	✓	✓

Table A12: Excluding Countries that were Ever Planned

	Outcome: Log GDP per capita						
	(1) CGV	(2) ANRR	(3) GWF	(4) GWF	(5) Polity	(6) V-Dem	(7) CGV
Autocracy	−0.101	−0.252	0.593	0.239	−0.121	0.127	0.082
	(0.326)	(0.306)	(0.424)	(0.313)	(0.334)	(0.319)	(0.295)
GS power consolidation	−0.717**	−0.746**					
	(0.360)	(0.361)					
GWF Categorical Personalism			−1.024*				
			(0.554)				
GWF Continuous Personalism				−0.394			
				(0.360)			
Executive Constraints					−0.131		
					(0.429)		
V-Dem Presidentialism						−0.871***	
						(0.305)	
Veto Players							−0.976***
							(0.323)
Autocracy + Personalism	−0.819***	−0.997***	−0.431	−0.155	−0.252	−0.745**	−0.895***
	(0.291)	(0.294)	(0.306)	(0.281)	(0.323)	(0.295)	(0.316)
p-value, unit root test	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Observations	5193	5386	4535	4535	4823	5266	5010
Countries	146	147	114	114	124	134	143
Country & year FE	✓	✓	✓	✓	✓	✓	✓
Four GDP lags	✓	✓	✓	✓	✓	✓	✓

Table A13: Controlling for Leader Changes

	Outcome: Log GDP per capita						
	(1) CGV	(2) ANRR	(3) GWF	(4) GWF	(5) Polity	(6) V-Dem	(7) CGV
Autocracy	−0.269	−0.351	0.415	0.023	−0.049	0.179	−0.204
	(0.390)	(0.363)	(0.451)	(0.359)	(0.317)	(0.331)	(0.354)
GS power consolidation	−0.802**	−0.918**					
	(0.371)	(0.407)					
GWF Categorical Personalism			−0.945*				
			(0.547)				
GWF Continuous Personalism				−0.201			
				(0.398)			
Executive Constraints					−0.104		
					(0.392)		
V-Dem Presidentialism						−0.858***	
						(0.283)	
Veto Players							−0.876**
							(0.341)
Leader Change	−1.239***	−1.202***	−0.481**	−0.476**	−0.947***	−1.306***	−1.243***
	(0.220)	(0.205)	(0.186)	(0.185)	(0.181)	(0.237)	(0.224)
Autocracy + Personalism	−1.071***	−1.269***	−0.530	−0.177	−0.153	−0.679**	−1.080***
	(0.321)	(0.366)	(0.324)	(0.317)	(0.317)	(0.311)	(0.352)
p-value, unit root test	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Observations	5710	5940	5104	5104	5395	5858	5516
Countries	176	178	144	144	154	165	172
Country & year FE	✓	✓	✓	✓	✓	✓	✓
Four GDP lags	✓	✓	✓	✓	✓	✓	✓

Table A14: Mechanisms

	(1) Hyperinflation	(2) Index of Market Reforms	(3) Log Trade	(4) Log tax to GDP
CGV Autocracy	−0.002	−0.633	−0.113	−2.920*
	(0.008)	(0.518)	(0.925)	(1.750)
GS Power Consolidation	0.001	0.578	−0.885	−0.425
	(0.005)	(0.358)	(0.747)	(1.474)
Autocracy + Personalism	−0.001	−0.056	−0.998	−3.344*
	(0.007)	(0.386)	(0.763)	(1.721)
Mean of Outcome Variable	0.014	42.575	411.849	-184.498
Observations	5710	4414	5167	4409
Countries	176	150	173	132
Country & year FE	✓	✓	✓	✓
Four outcome lags	✓	✓	✓	✓
Four GDP lags	✓	✓	✓	✓

Table A15: Annual transitions among regime types

		CGV–GS$_t$		
		Democracy	Institutionalized	Personalist
CGV–GS$_{t-1}$	Democracy	2,999	7	45
	Institutionalized	21	1,711	61
	Personalist	59	50	2,340

		GWF–GWF$_t$		
		Democracy	Institutionalized	Personalist
GWF–GWF$_{t-1}$	Democracy	2,544	27	28
	Institutionalized	43	1,966	22
	Personalist	24	17	1,656